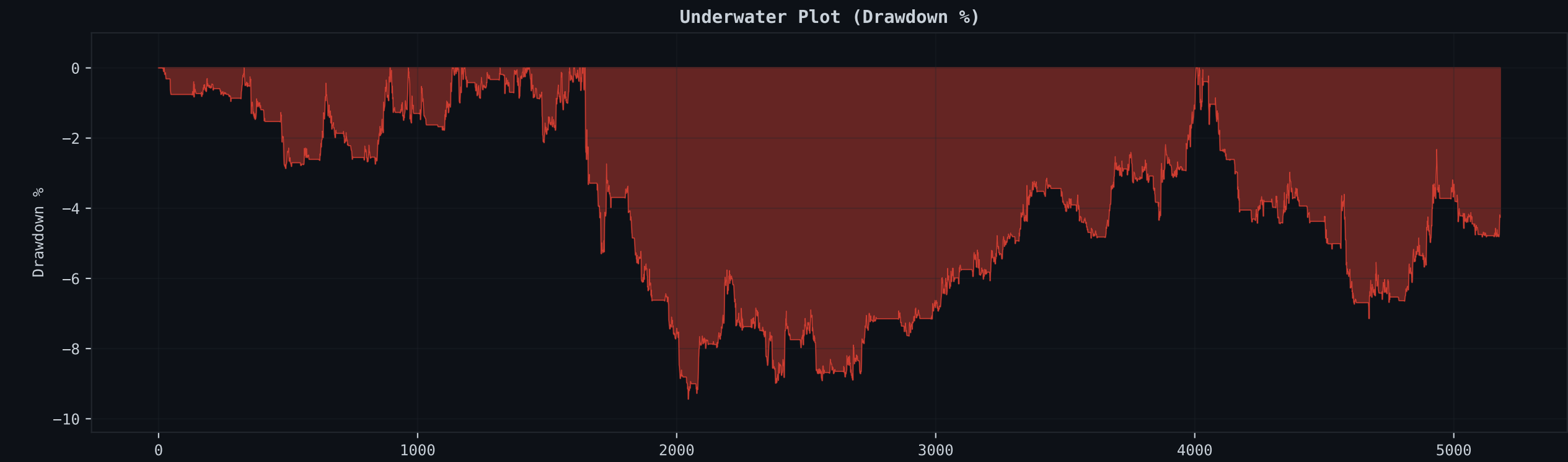
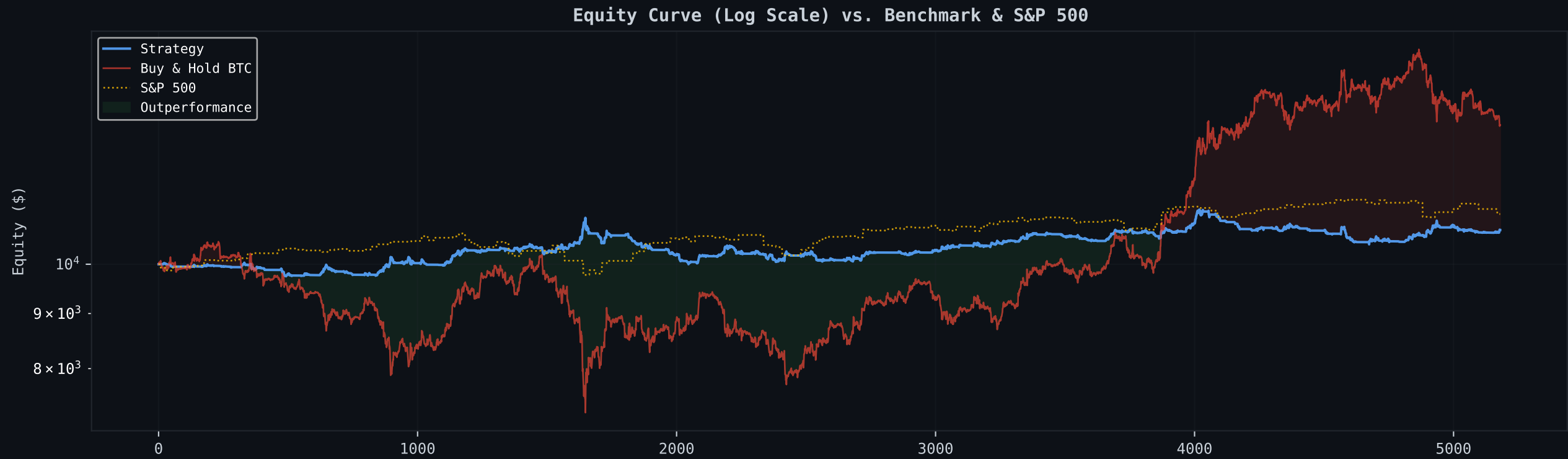
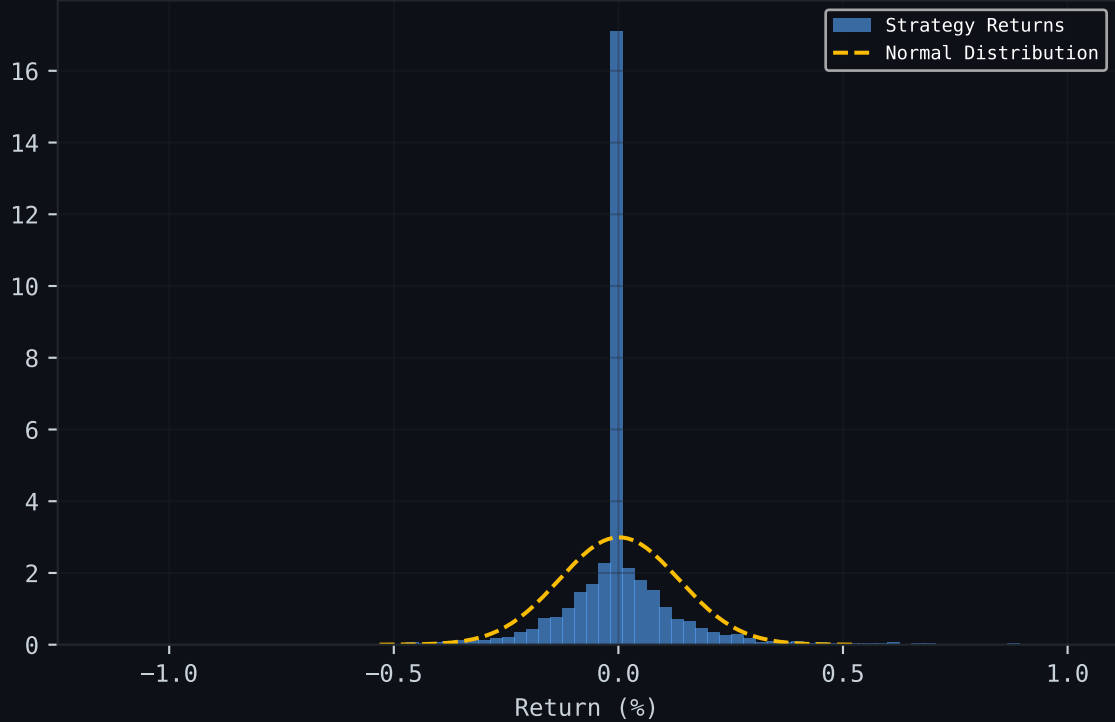


INSTITUTIONAL QUANTITATIVE TEARSHEET
CNN+LSTM PPO × SPA Day Trading Strategy

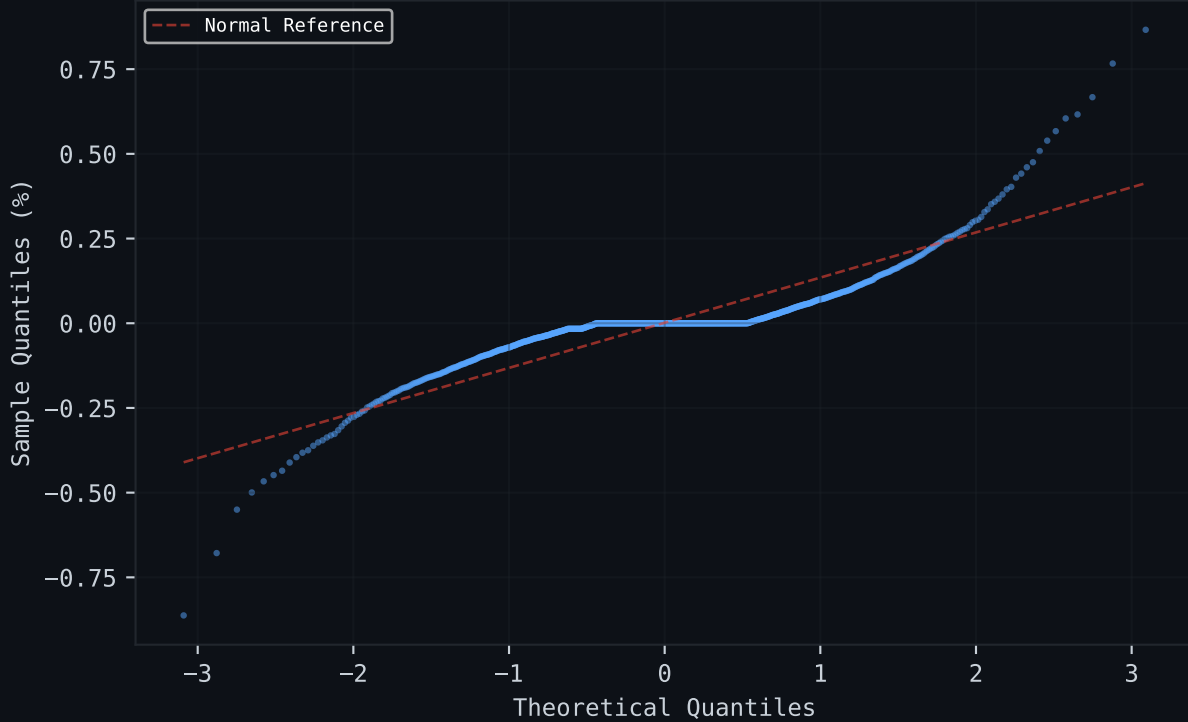
Sharpe: 1.05 | Sortino: 1.18 | Calmar: 1.39 | Max DD: -9.4% | Alpha p-value: 0.2034 | Beta: -0.024



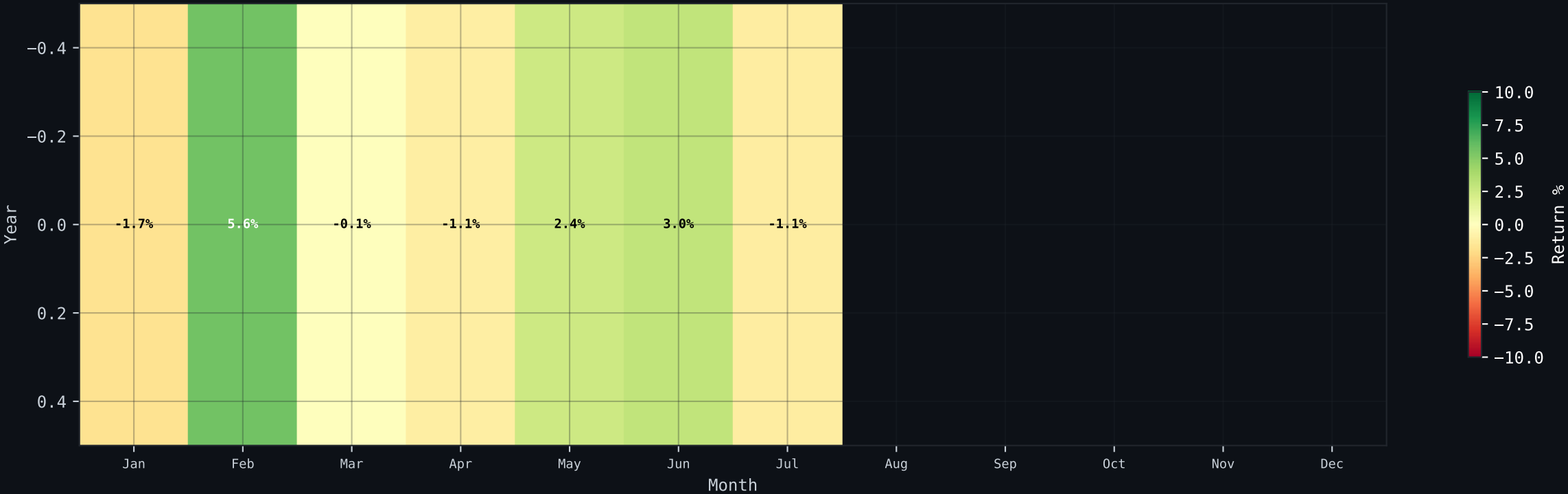
Return Distribution vs. Normal



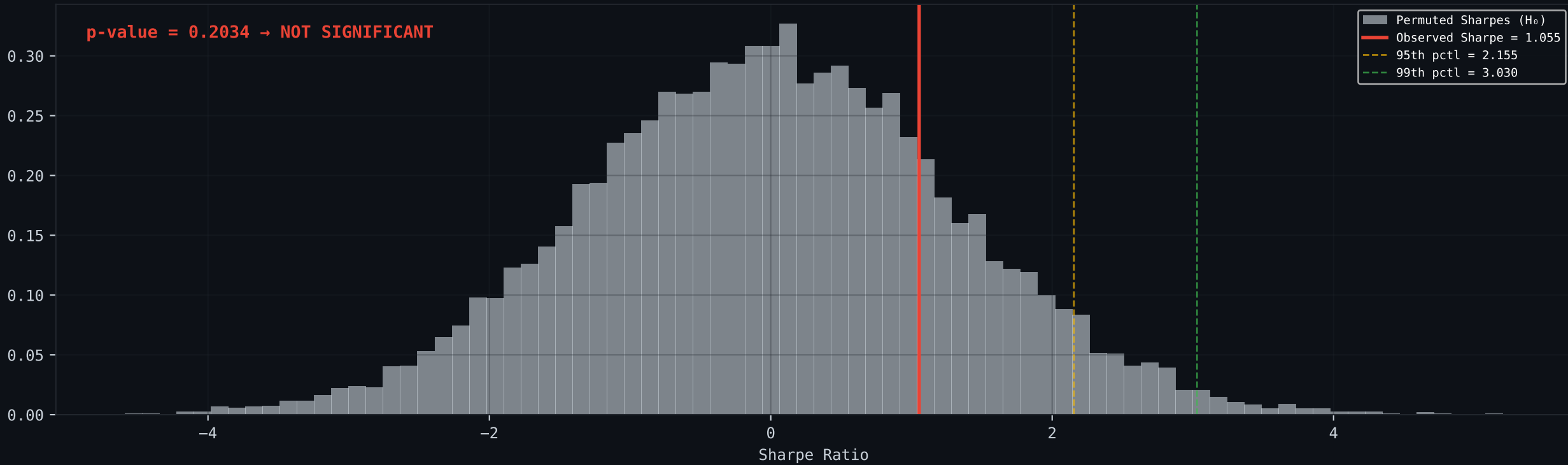
Q-Q Plot (Fat Tail Detection)



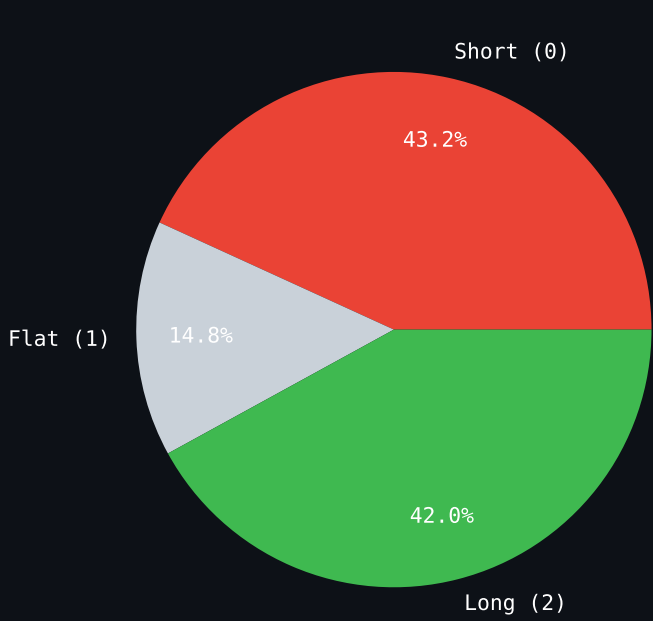
Monthly Returns Heatmap (%)



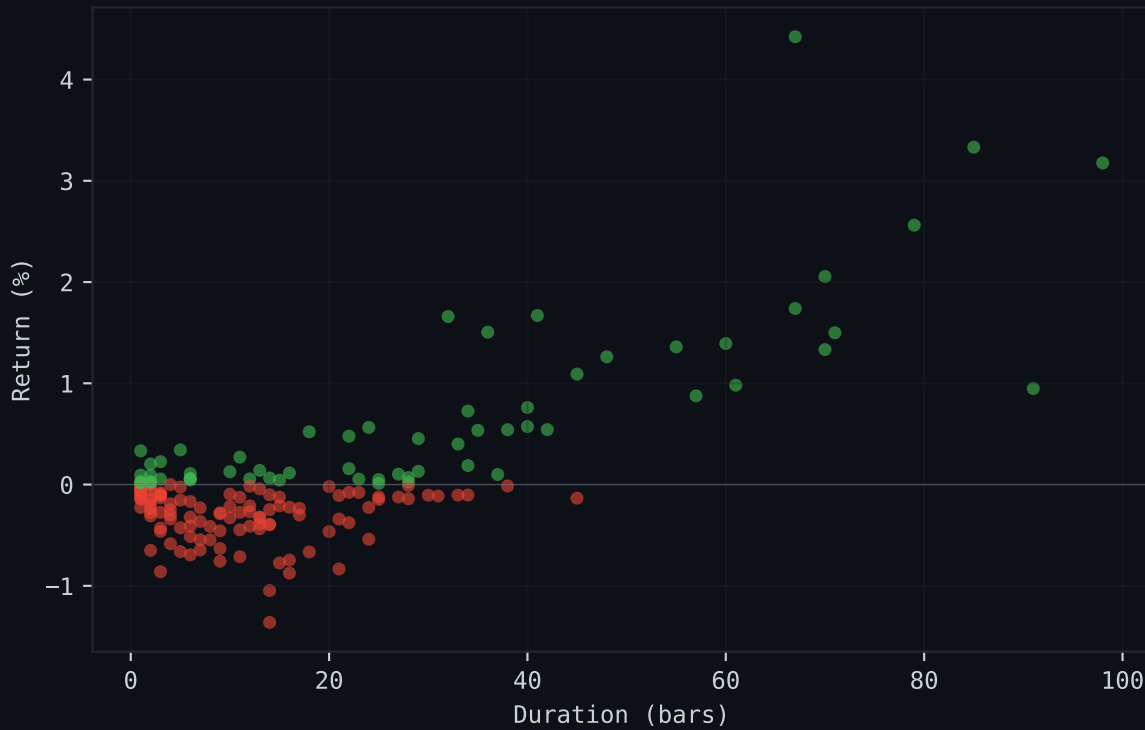
Monte Carlo Permutation Test (10,000 simulations)



Action Distribution



Trade Returns vs. Duration



QUANTITATIVE RISK METRICS – FULL DETAIL

Metric	Value
PERFORMANCE METRICS	
Total Return	7.58%
Annualized Return	13.16%
Annualized Volatility	12.48%
Sharpe Ratio	1.055
Sortino Ratio	1.183
Calmar Ratio	1.394
Omega Ratio (θ=0)	1.044
Profit Factor	1.044
Win Rate	47.3%
BENCHMARK COMPARISON	
BTC Buy & Hold Sharpe	1.294
BTC Buy & Hold Sortino	1.706
BTC Buy & Hold Max DD	-30.61%
S&P 500 Sharpe	1.468
S&P 500 Sortino	0.305
S&P 500 Max DD	-8.49%
TAIL RISK METRICS	
Max Drawdown	-9.44%
VaR (95%)	-0.1865%
VaR (99%)	-0.3821%
CVaR / Expected Shortfall (95%)	-0.3168%
CVaR / Expected Shortfall (99%)	-0.5577%
Skewness	0.260
Excess Kurtosis	12.875
ALPHA DECOMPOSITION	
Alpha (per period)	0.000017
Alpha t-statistic	0.908
Alpha p-value	0.3641
Beta (benchmark exposure)	-0.0241
R² (benchmark correlation)	0.0095
STATISTICAL VALIDATION	
Permutation Test p-value	0.2034
Significance (p < 0.05)	NO ☐
CAPACITY ESTIMATION	
Est. Capacity (USD)	\$11,123,756,362,760
Current Market Impact	0.33 bps
Participation Rate	0.000011%